RELATIVE MOTION DAY 2

LEARNING GOALS

Students will explore various aspects of relative motion

- Be able to draw relative vector equations and related diagrams

CONCEPT QUESTIONS

1. Car 1 is driving at 45 km/h [W] with respect to the road. Car 2 is driving at 52 km/h [E] with respect to the road. What is the velocity of car 2 with respect to car 1? [97 km/h [E]]

$\vec{V}_{1G} = 45\,km/h\,[W]$ ⟵

⟶ $\vec{V}_{2G} = 52\,km/h\,[E]$

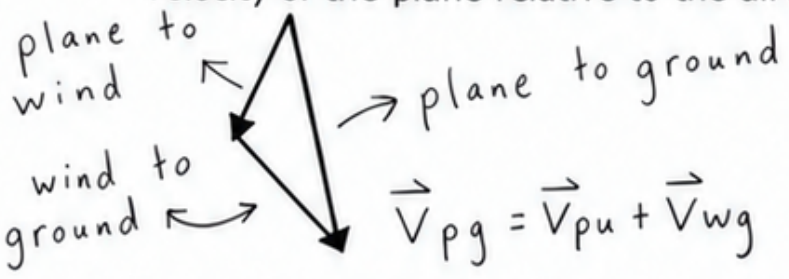
$\vec{V}_{21} = \vec{V}_{2G} - \vec{V}_{1G}$

$\vec{V}_{21} = 52 - (-45)$ (→) W ⟷ E (+)

$v_{21} = +97\,km/h$

$\boxed{V_{21} = 97\,km/h\,[E]}$

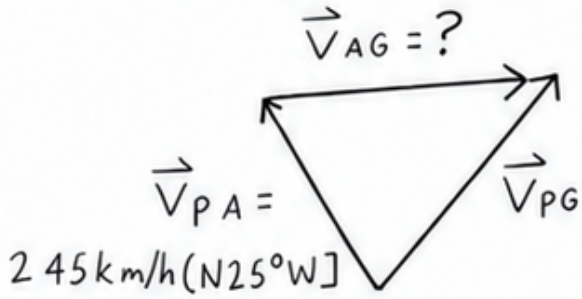
2. If the wind is represented by the arrow pointing south-east, which arrow represents the velocity of the plane relative to the air?



the arrow pointing south west represents the velocity of the plane relative to the air.

3. A pilot aims [N25°W] with an airspeed of 245 km/h in order to achieve the groundspeed shown below. Draw a vector diagram to show the direction of the wind.





$\vec{V}_{AG} = \vec{V}_{AP} + \vec{V}_{PG}$

$= (-\vec{V}_{PA}) + \vec{V}_{PG}$

4. A football is thrown through the air at 6.0 m/s [S], but a quarterback running down the field sees it moving at 2.5 m/s [N]. Determine the speed and direction the quarterback is running.

⟵ $\vec{V}_{FA} = 6.0\,m/s\,[S]$

⟶ $\vec{V}_{FQ} = 2.5\,m/s\,[N]$

⟵ $\vec{V}_{QA} = \vec{V}_{FA} + \vec{V}_{QF}$

$\vec{V}_{QA} = \vec{V}_{FA} + (-\vec{V}_{FQ})$

$\vec{V}_{QA} = -6 - 2.5$

$\boxed{\vec{V}_{QA} = -8.5\,m/s}$ S (-) ⟷ N (+)

$\vec{V}_{QA} = 8.5\,m/s\,[S]$

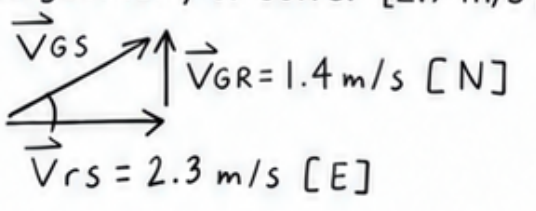
PROBLEM SOLVING

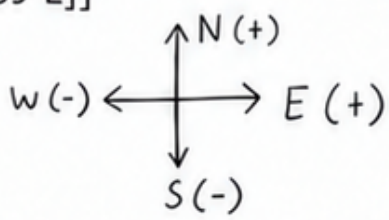
EXAMPLE ONE: RIVER CURRENT

A girl tries swimming across a river. The river is flowing at 2.3 m/s [E]. The girl swims at 1.4 m/s [N] with respect to the water. What is the girl's velocity with respect to the river bank (the shore)? Draw a labelled vector diagram and use geometry to solve. [2.7 m/s [N59°E]]

$\vec{V}_{GS}$

$\vec{V}_{GS} = \vec{V}_{rs} + \vec{V}_{GR}$ ✓ ✓





$v_{GS}^2 = v_{GR}^2 + v_{rs}^2$

$v_{GS} = 2.69\,m/s$ ✓

$\theta = \tan^{-1}\left(\frac{v_{GR}}{v_{rs}}\right)$

$\therefore \vec{V}_{GS} = 2.7\,m/s\,[E\,31°\,N]$ ✓

$\theta = 31°$ north of east ✓

EXAMPLE TWO: PLANE

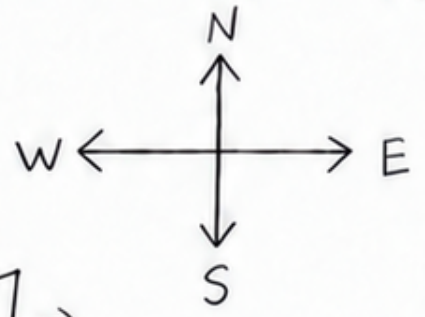
A plane is flying with an airspeed of 340 km/h and a heading of west. The plane's velocity over the ground is 365 km/h [W16°S]. What is the wind velocity? Draw a labelled vector diagram and use geometry or components to solve. [101 km/h [S 6° W]

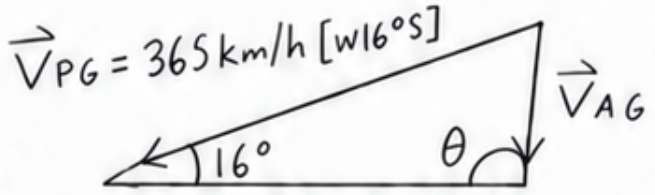
$\vec{V}_{PA}$ $\vec{V}_{PG}$ $\vec{V}_{AG}$

$$\vec{V}_{PG} = \vec{V}_{PA} + \vec{V}_{AG}$$

$$\vec{V}_{AG} = \vec{V}_{PG} - \vec{V}_{PA}$$

$$\vec{V}_{AG} = \vec{V}_{PG} + (-\vec{V}_{PA})$$





$$-\vec{V}_{PA} = 340 \text{ km/h [E]}$$

$$\frac{\sin\theta}{365} = \frac{\sin 16°}{101.2}$$

$$\theta = 96° \text{ (triangle angle)}$$

$$V_{AG}^2 = V_{PG}^2 + V_{PA}^2 - 2(V_{PG})(V_{PA})\cos 16°$$

$$V_{AG}^2 = 10239.85$$

$$V_{AG} = 101.2 \text{ km/h}$$

$$\vec{V}_{AG} = 101 \text{ km/h [S 6° W]}$$

PROBLEMS TO SOLVE

1. A boat crosses a river of width 160 m in which the current has a uniform speed of 1.50 m/s. The pilot maintains a bearing perpendicular to the river and a throttle setting to give a constant speed of 2.00 m/s relative to the water. (a) What is the speed of the boat relative to a stationary shore observer? (b) How far downstream from the initial position is the boat when it reaches the opposite shore? [(a) 2.5 m/s [N36.9°E (b) 120 m]

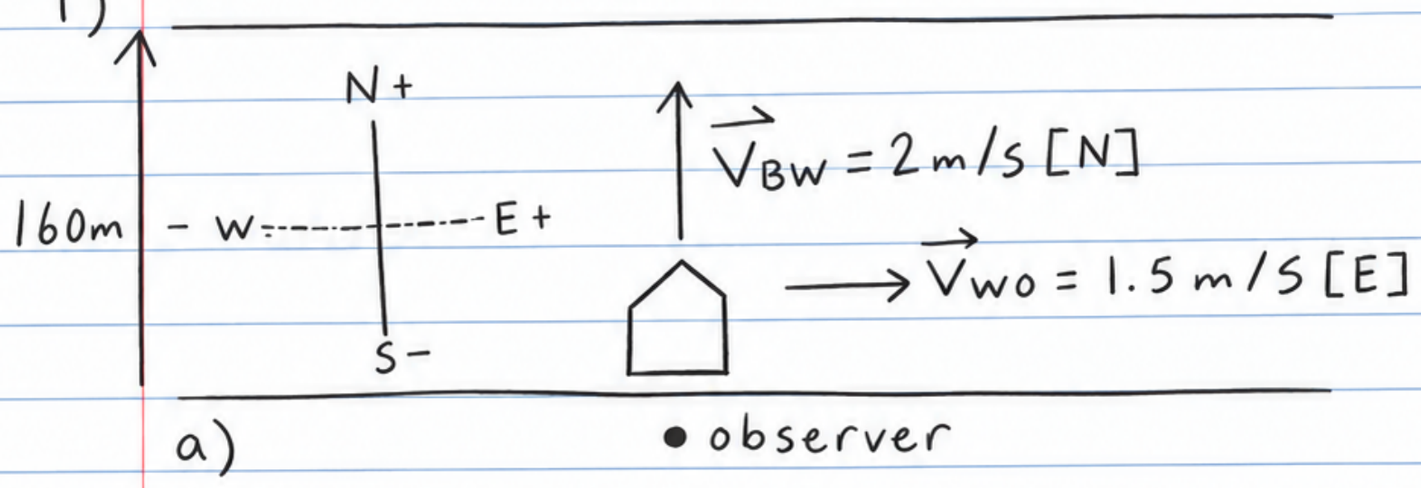
2. A river has a steady speed of 0.500 m/s. A student swims upstream a distance of 1.00 km and the swims back to the original starting point. If the student can swim at a speed of 1.20 m/s in still water, how long does the trip take? Compare this with the time the trip would take if the water were still. [2.0×10^3 s no current: 1.67×10^3 s]

3. The pilot of an airplane notes that the compass indicates a heading due west. The plane's airspeed (speed relative to the air) is 150 km/h. If there is a wind of 30.0 km/h toward the north, find the velocity of the plane with respect to the ground. [153 km/h [W11.3°N]]

4. Two planes fly from Toronto to Philadelphia. Plane A flies via Pittsburgh whereas passengers on plane B have a direct flight. Pittsburgh is 350 km due south of Toronto and 390 km due west of Philadelphia. The airspeed of both planes is 400.0 km/h and a steady wind is blowing from the east at 60.0 km/h. (a) What direction must the pilot point the plane flying from Toronto to Pittsburgh? Include a vector diagram of velocities. (b) How long will the entire flight take for plane A assuming a 0.50-h layover in Pittsburgh? (c) How much time must the pilot of plane B wait before leaving Toronto if she is to arrive in Philadelphia at the same time plane A arrives? [(a) 8.6° (b) v_{PB} = 353 km/h (b) 2.53 h (c) waits 1.0 h]

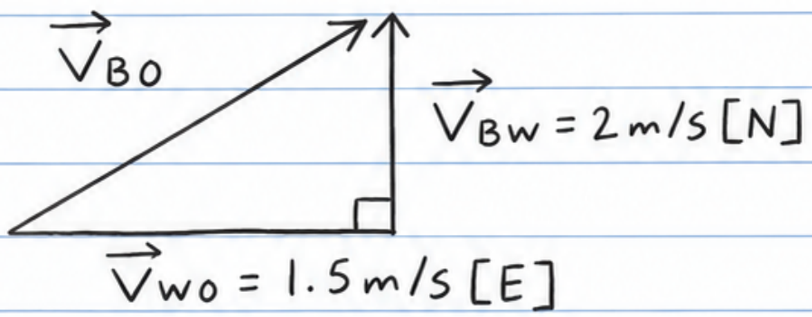
1.3 b HW (relative motion part 2)

1)



a)

$$\vec{V}_{BO} = \vec{V}_{WO} + \vec{V}_{BW}$$



$$V_{BO}^2 = V_{WO}^2 + V_{BW}^2$$
$$V_{BO}^2 = 1.5^2 + 2^2$$
$$V_{BO} = \sqrt{1.5^2 + 2^2}$$
$$\boxed{V_{BO} = 2.5\,m/s}$$

b) $d_x = V_x \Delta t \quad d_{Bx} = V_{WO} \Delta t$

$V_{WO} = 1.5\,m/s$

find Δt based on V_{BW} and river width (160m)

$d = v\Delta t$

$160 = V_{BW} \Delta t$

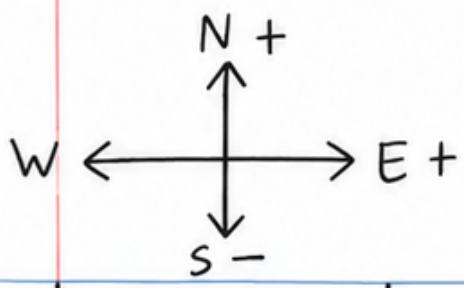
$160 = 2\Delta t$

$\boxed{80 = \Delta t}$

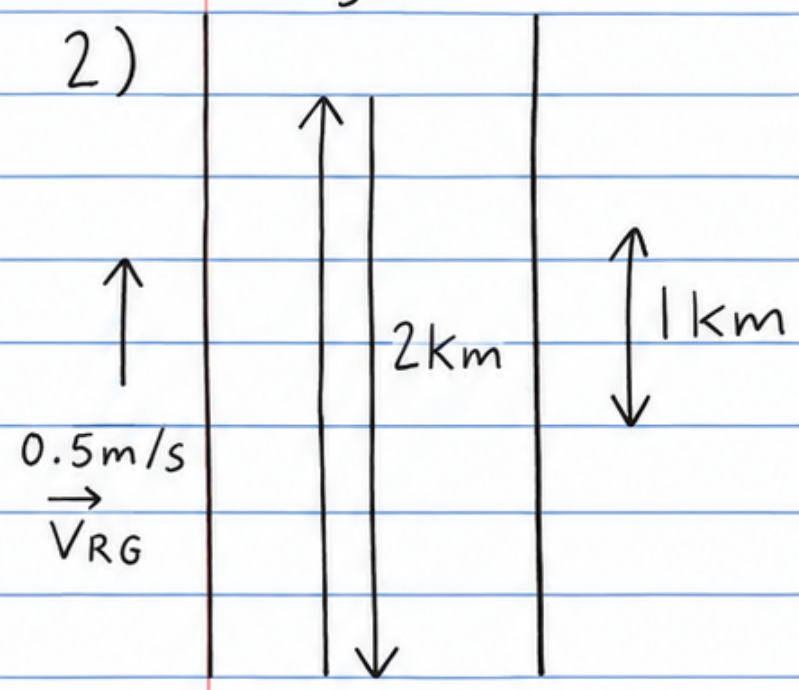
$d_{Bx} = V_{WO} \Delta t$

$d_{Bx} = 1.5 \times 80$

$\boxed{d_{Bx} = 120\,m}$



2)



In still water

$V_{SG} = 1.2 m/s$

find student's velocity downstream

$\vec{V}_{SG} = \vec{V}_{SR} + \vec{V}_{RG}$

$\vec{V}_{SG} = 1.2 + 0.5$

$\vec{V}_{SG} = 1.7 m/s\ [N]$

find time

$\Delta t = \Delta d / \vec{v}$ $\Delta t = 1000 / 1.7$

$\Delta t = 588 s$

find student's velocity upstream

$\vec{V}_{SG} = \vec{V}_{SR} + \vec{V}_{RG}$

$\vec{V}_{SG} = -1.2 + 0.5$

$\vec{V}_{SG} = 0.7 m/s\ [S]$

find time

$\Delta t = \Delta d / \vec{v}$ $\Delta t = 1000 / 0.7$

$\Delta t = 1429 s$

total time = 2017s

find Δt if the water was still

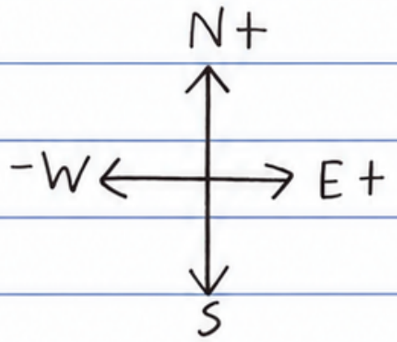
$d = 2000\,m.$ $V = 1.2\,m/s$

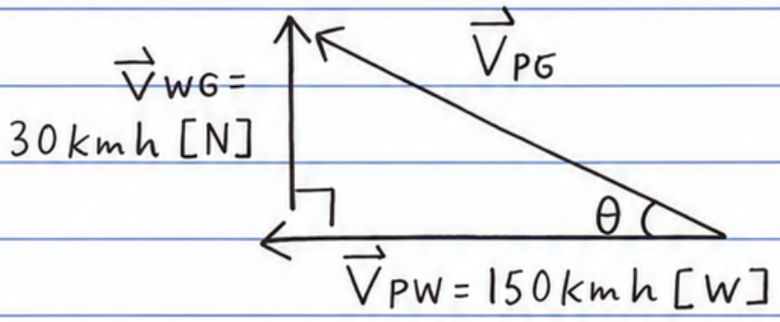
$\Delta t = d/v$ $\Delta t = 2000/1.2$

$\Delta t = 1667\,s$

$\therefore$ the current makes the trip take 350 s longer

3)





$\vec{V}_{PG} = \vec{V}_{PW} + \vec{V}_{WG}$

$V_{PG} = \sqrt{V_{PW}^2 + V_{WG}^2}$

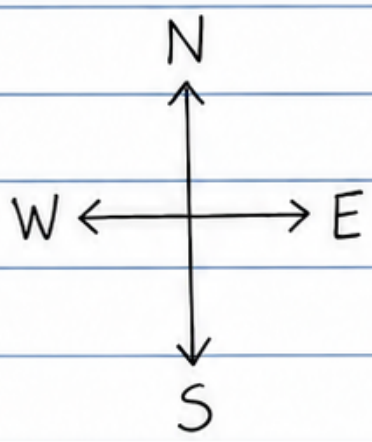
$V_{PG} = 153\,km/h$

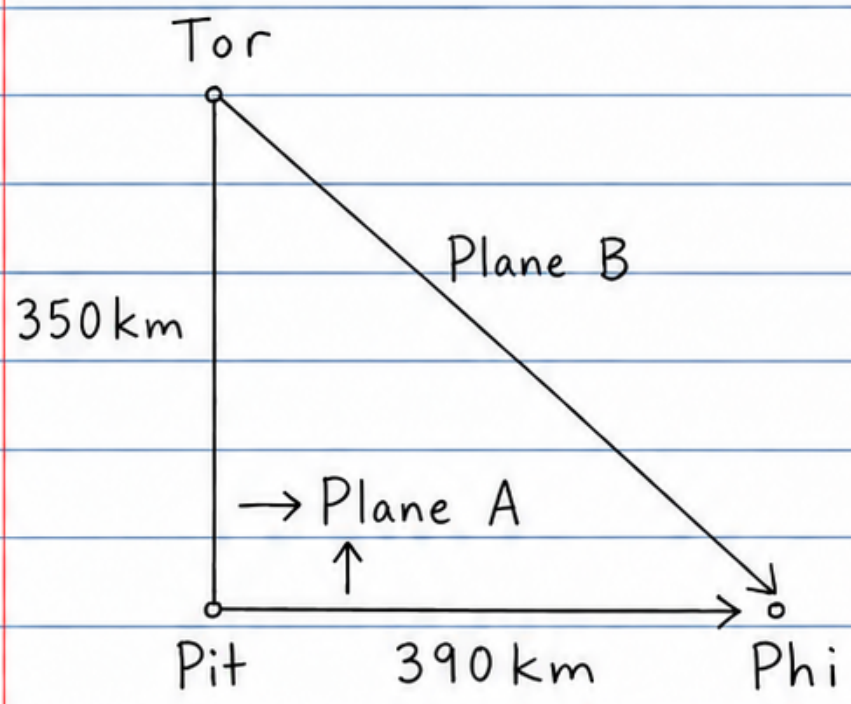
$\theta = \tan^{-1}(V_{WG}/V_{PW})$

$\theta = 11.3°$ north of west

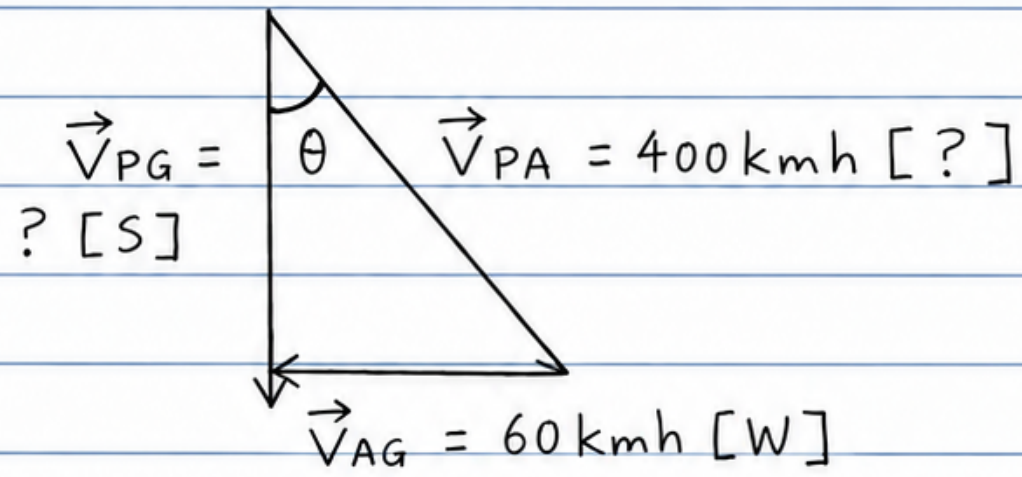
$\therefore \vec{V}_{PG} = 153\,km/h\ [W\,11.3°N]$

4)





a)



$\sin\theta = V_{AG} / V_{PA}$

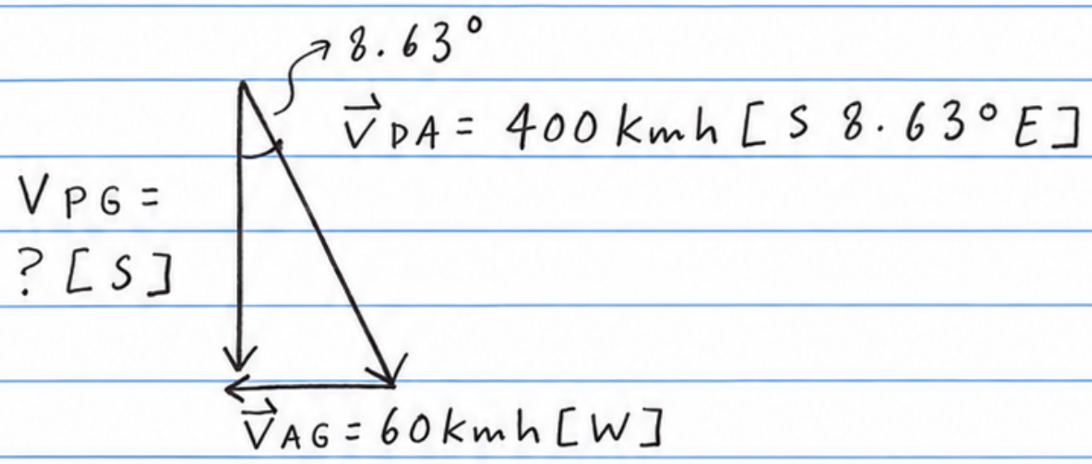
$\theta = \sin^{-1}(60/400)$

$\theta = 8.63°$ east of south

$\therefore \vec{V}_{PA} = 400\text{ kmh } [S\ 8.63°E]$

$\therefore$ the pilot should aim [S 8.63° E]

b) Toronto to pittsburgh



use $a^2 + 6^2 = c^2$

$V_{PG}^2 + 60^2 = 400^2$

$V_{PG} = \sqrt{400^2 - 60^2}$

$V_{PG} = 395$ kmh

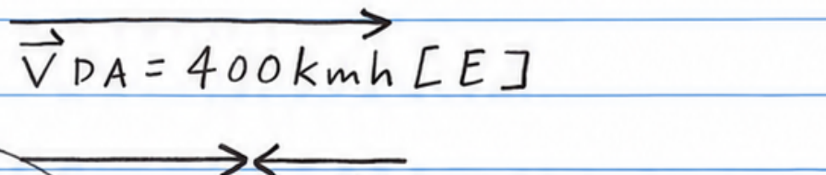
$\vec{V}_{PG} = 395$ kmh [S]

$\Delta t = \Delta d / \vec{V}_{PG}$

$\Delta t = 350 / 395$

$\Delta t = 0.89$ hours

pittsburgh to philadelphia



$\vec{V}_{PG} = \vec{V}_{DA} + \vec{V}_{AG}$

$V_{PG} = 400 + (-60)$

$\vec{V}_{PG} = +340$ kmh

$\vec{V}_{PG} = 340$ kmh [E]

$\Delta t = \Delta d / \vec{V}_{PG}$ $\Delta t = 390 / 340$

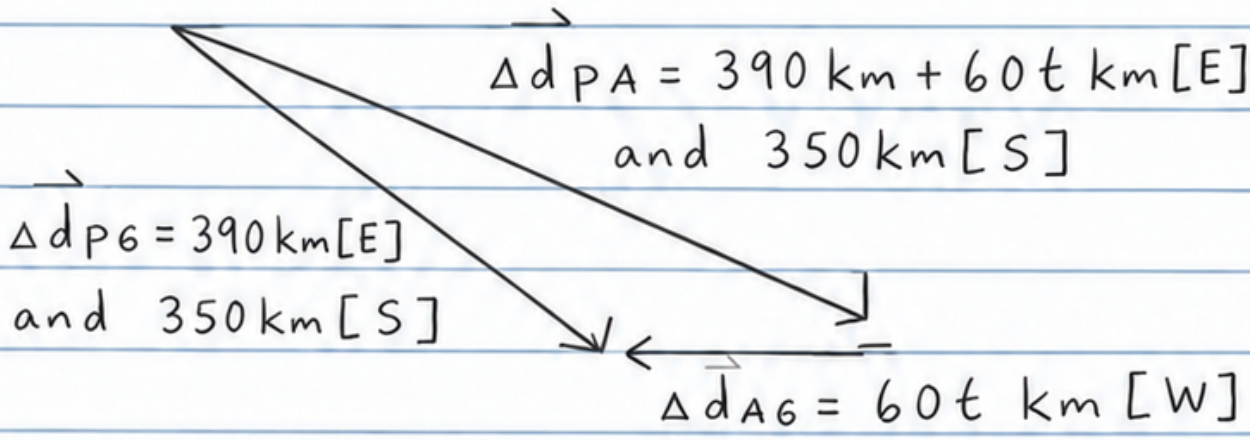
$\Delta t = 1.15$ hours

$\Delta t_{total} = \Delta t_1 + \Delta t_2 + 0.5\,h$ (layover)

$\Delta t_{total} = 0.89h + 1.15h + 0.5h$

$\Delta t_{total} = 2.54\ h$

Find Δt for direct flight?

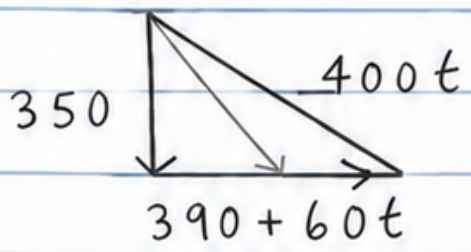


$\Delta d_{PA} = 400t$

$\Delta d_{PAx} = 390 + 60t$

$\Delta d_{PAy} = -350\,km$

$V_A = 60\,km/h\ [W]$



use $a^2 + b^2 = c^2$

$(390 + 60t)^2 + 350^2 = (400t)^2$

$152100 + 46800t + 3600t^2 + 122500 = 160000t^2$

$156400t^2 - 46800t - 274600 = 0$

use quadratic formula

$\Delta t_2 = 1.48\ h$ (direct)

$\Delta t_1 = 2.54$ (layover)

wait $= \Delta t_1 - \Delta t_2$

wait $= 1.06\ h$

$\therefore$ plane B must wait 1.06 h

The rest of the answers are on the digital copy